

NSCLC Clinical Trials

Investment & Risk Assessment

A due-diligence-style analysis of the Non-Small Cell Lung Cancer (NSCLC) drug development landscape

Prepared using SQL, Excel, and Power BI analysis of the AACT clinical trials database

How to Read This Report

This report looks at the market for new lung cancer drugs specifically for a type of lung cancer called Non-Small Cell Lung Cancer, or "NSCLC" for short (about 85% of all lung cancer cases are this type). No prior background in biotech, medicine, or finance is needed to follow it. Anywhere a technical term first appears, it's explained in plain language.

Think of drug development like a series of exams a new medicine has to pass before it's allowed to be sold to patients:

- Phase 1: Is it safe? (small group of people, checking for obvious harm)
- Phase 2: Does it seem to work? (larger group, looking for early signs of benefit)
- Phase 3: Does it really work, at scale? (large group, the main test regulators look at)
- Phase 4: Ongoing checks after the drug is already approved and being sold

Most drugs fail somewhere along this path that's normal, and a central theme of this report is exactly how much risk exists at each stage, and what that risk means commercially.

Executive Summary

This report analyzes real clinical trial data for NSCLC to answer a practical business question: where is drug development activity concentrated, how risky is it at each stage, who is winning, and how big is the financial opportunity for a company competing in this space?

Bottom line: NSCLC drug development is a high-risk, high-reward field. Only about 5% of drugs that start human testing for cancer ever reach approval. But for a drug that succeeds like AstraZeneca's Tagrisso the market opportunity is worth billions of dollars per year.

Key findings:

- Development activity is heavily concentrated in Phase 2 (the "does it work" stage) more trials happen here than in any other phase.
- Risk drops steadily as a drug moves through the phases from a 21.5% chance of a trial being stopped in Phase 1, down to just 2% by Phase 4.
- Early research is led by universities and research hospitals (especially the National Cancer Institute); large-scale, late-stage trials are led by major pharmaceutical companies chiefly AstraZeneca and Hoffmann-La Roche, who are the only two companies with a presence across every stage of development.
- For AstraZeneca's leading NSCLC drug, Tagrisso, the realistic yearly market opportunity is estimated at roughly \$3.7 billion out of a much larger theoretical market of \$54.5 billion.
- Despite Tagrisso's strong position, the wider NSCLC market is highly competitive, with several major companies racing to serve overlapping groups of patients.

Data & Methodology

All findings in this report are based on real, publicly available clinical trial records for NSCLC, pulled from AACT (Aggregate Analysis of ClinicalTrials.gov) a free database that mirrors the official U.S. government clinical trials registry. Analysis was performed using SQL (to query the trial database directly), Excel (to build financial models), and Power BI (to build the visual dashboard referenced throughout this report).

Where this report uses estimates or industry benchmarks rather than directly observed data, that is stated explicitly, along with the source.

1. Where Is Development Activity Concentrated?

Looking across roughly 3,400 NSCLC clinical trials, activity is not evenly spread across development stages. The chart below (see accompanying Power BI dashboard, "Where Is NSCLC Trial Activity Concentrated?") shows trial counts by phase:

Phase	Number of Trials	What This Phase Means
Phase 2	1,130	Largest single stage most drugs that pass initial safety testing are tested here for early signs of benefit
Phase 1	590	Initial safety testing
Phase 3	518	Large-scale confirmation testing
Phase 1/2	418	Combined safety + early benefit testing
Phase 2/3	58	Combined early + large-scale testing
Phase 4	50	Post-approval monitoring

Why this matters: Phase 2 is the stage where most exploratory research concentrates, because it's the first real test of whether a drug actually helps patients as a result, it naturally attracts the largest number of competing drug candidates, and is where the most is generally at stake.

2. How Risky Is Each Stage?

Not every clinical trial that starts finishes successfully many are stopped early ("terminated") for reasons like poor early results, funding issues, or safety concerns. Looking at how often trials get terminated at each phase gives a real, data-backed picture of risk:

Phase	Termination Rate	What This Means
Phase 1	21.5%	Highest risk first time in humans
Early Phase 1	18.8%	Very early testing
Phase 1/2	16.8%	Still early-stage risk
Phase 2/3	12.1%	Risk narrowing

Phase	Termination Rate	What This Means
Phase 2	11.5%	Risk narrowing
Phase 3	8.1%	Lower risk drug has already proven itself earlier
Phase 4	2%	Lowest risk drug is already approved and selling

In plain terms: the further a drug advances, the more "proven" it has become, and the less likely it is to fail. This is exactly what you'd expect and it matches how the wider pharmaceutical industry generally works.

A Deeper Risk Check: How Likely Is Approval, Really?

A trial not being "terminated" isn't the same as a drug actually succeeding a trial can run to completion and still fail to prove the drug works well enough for approval. To capture this more accurately, this report also compares two numbers:

- A simple estimate based only on this dataset's termination rates: roughly 72% (this is an oversimplified, illustrative figure it only accounts for the risk of a trial being stopped, not the risk of a completed trial still failing)
- The real, industry-reported likelihood of a cancer drug reaching approval after starting Phase 1 human testing: only about 5% (source: BIO the Biotechnology Innovation Organization via Informa Biomedtracker data)

The gap between these two numbers (72% vs. 5%) is itself an important finding: it shows that simply avoiding early termination is not enough for a drug to succeed most of the real risk in cancer drug development comes from trials that run their full course but still fail to prove the drug works well enough. For comparison, non-oncology drugs succeed from Phase 1 to approval at a notably higher rate of around 12% reflecting how uniquely difficult cancer drug development is.

3. Who Is Leading NSCLC Development?

Looking at which organizations run the most NSCLC trials reveals a clear pattern: research and commercial activity are led by different types of players at different stages.

Organization	Total Trials	Stage Focus	Type
National Cancer Institute	93	Early & Mid-Stage	Government research institute
Hoffmann-La Roche	90	Full Pipeline (heavy late-stage)	Pharmaceutical company
AstraZeneca	62	Full Pipeline	Pharmaceutical company
Eli Lilly and Company	61	Mixed	Pharmaceutical company
Sun Yat-sen University	56	Early-Stage	Academic institution

Only two organizations AstraZeneca and Hoffmann-La Roche run trials across the entire development spectrum, from the earliest safety testing through to post-approval monitoring. This "full pipeline" presence is a meaningful sign of commercial strength: it means a company isn't relying on a single drug, but continuously feeding new candidates into its pipeline while also defending and expanding its already-approved products.

By contrast, the National Cancer Institute despite running the most trials overall is concentrated almost entirely in early and transitional stages (particularly the combined Phase 1/2 category), consistent with its role as a public research funder rather than a commercial drug developer.

4. How Big Is the Financial Opportunity?

To estimate the size of the commercial opportunity, this report focuses on one specific, well-known NSCLC drug AstraZeneca's Tagrisso and works through three standard tiers of market sizing, from broadest to most realistic:

- TAM (Total Addressable Market): the value of the market if every single NSCLC patient in the US were treated a theoretical ceiling, not a realistic target
- SAM (Serviceable Addressable Market): the value of the market Tagrisso could realistically address only patients whose cancer has the specific genetic mutation ("EGFR") that Tagrisso targets, about 15% of NSCLC patients
- SOM (Serviceable Obtainable Market): a realistic estimate of what AstraZeneca could actually capture, accounting for how many eligible patients get diagnosed and treated, and how much of that treated group chooses Tagrisso over competing drugs

Market Tier	Patients per Year	Estimated Annual Value
TAM	~195,000 patients/year	\$54.5 billion
SAM	~29,250 patients/year	\$8.2 billion
SOM	~13,300 patients/year	\$3.7 billion

Sources: US lung cancer case counts from the American Cancer Society (Cancer Facts & Figures 2026); EGFR mutation prevalence from a 2026 peer-reviewed review in the Journal of the National Cancer Institute (JNCI); treatment-rate and market-share figures are reasonable modeling assumptions, stated explicitly as such rather than presented as hard data.

In plain terms: even after narrowing down to only the patients Tagrisso is actually designed for, and accounting for real-world competition, the drug's realistic yearly opportunity is still around \$3.7 billion a substantial, attractive market by any standard.

5. Competitive Forces Shaping This Market

Using a well-known business framework (Porter's Five Forces), here's what shapes competition in the NSCLC drug market:

Threat of New Entrants: Moderate-Low

It's hard for new companies to break into this space developing a cancer drug is enormously expensive and slow, and (as shown above) only about 5% of candidates that start testing ever get approved. Most new activity starts in universities and research institutions rather than brand-new companies.

Bargaining Power of Suppliers: Low-Moderate

"Suppliers" here mainly means the companies that run clinical trials on behalf of drugmakers, and manufacturing inputs. Large pharmaceutical companies typically have enough trial volume to negotiate favorably.

Bargaining Power of Buyers: High

"Buyers" here means insurers, governments, and patients who ultimately pay for these drugs. Prices vary enormously from about \$54,000 to \$279,000 per year across the ten major NSCLC drugs studied for this report and payers increasingly push back on high prices, especially once cheaper generic alternatives become available.

Threat of Substitutes: Moderate, Rising

Patients and doctors have real choices between different types of treatment (targeted therapies, immunotherapies, chemotherapy), and once a drug's patent protection ends, cheaper generic versions can take over much of its market this already happened to one of the ten drugs studied (Tarceva), whose brand-name version was discontinued entirely.

Competitive Rivalry: High

Multiple large pharmaceutical companies are simultaneously running full pipelines of competing drugs aimed at overlapping groups of patients this is a genuinely crowded, fast-moving market, not one dominated by a single winner.

6. Company Spotlight: AstraZeneca

Given AstraZeneca's full-pipeline presence and market-leading position with Tagrisso, this section takes a closer look specifically at AstraZeneca's strengths, weaknesses, opportunities, and threats within the NSCLC market.

Category	Analysis
Strengths	Full-pipeline presence (one of only two such companies); Tagrisso is the standard first-choice treatment for its target patient group; established regulatory track record since 2015
Weaknesses	Tagrisso only works for a specific genetic subtype of NSCLC (~15% of patients) a narrow addressable population; highest price among the ten drugs studied, increasing exposure to payer pushback
Opportunities	Meaningful gap between the realistically reachable market (SAM, \$8.2B) and what's currently captured (SOM, \$3.7B) room to grow through better patient diagnosis rates or gains from competitors; continued pipeline investment across all phases suggests future growth is being actively pursued
Threats	Intense competition from Roche, Bristol-Myers Squibb, Merck, and others targeting overlapping patient groups;

Category	Analysis
	eventual patent expiration and generic competition already seen with a competitor drug in this same study; ongoing pressure from insurers and governments on drug pricing

7. Conclusion & Recommendation

NSCLC drug development is a high-risk, high-reward field the overwhelming majority of drugs that enter testing will fail, and the ones that succeed do so only after clearing a demanding, expensive, multi-year path. But for the small number of drugs that do succeed, the commercial reward is substantial: a single successful drug like Tagrisso represents a realistic multi-billion-dollar annual opportunity.

For an investor, consultant, or company evaluating this space, three things stand out from this analysis:

- Development risk is front-loaded the earliest stages carry by far the highest chance of failure, meaning the most value (and the most uncertainty) sits with companies and institutions active in Phase 1 and Phase 2 research.
- Commercial success depends on full-pipeline strength, not a single drug AstraZeneca and Roche's presence across every development stage, rather than reliance on one product, is what distinguishes genuine market leaders from one-off successes.
- Even a narrow, biomarker-defined market can be highly valuable Tagrisso addresses only about 15% of NSCLC patients, yet still represents a multi-billion-dollar opportunity, underscoring how large and valuable the overall NSCLC market is.

Recommendation: Companies or investors evaluating opportunities in NSCLC should prioritize partners or targets with full-pipeline presence (like AstraZeneca or Roche) over single-drug plays, given the demonstrated importance of sustained, multi-phase pipeline investment in this market.

Appendix: Data Sources

- Clinical trial data: AACT (Aggregate Analysis of ClinicalTrials.gov), Clinical Trials Transformation Initiative
- US lung cancer incidence: American Cancer Society, Cancer Facts & Figures 2026
- EGFR mutation prevalence: Journal of the National Cancer Institute (JNCI), 2026 review article
- Oncology drug approval likelihood: BIO (Biotechnology Innovation Organization), via Informa Biomedtracker data, as reported in Applied Clinical Trials Online
- Drug pricing: GoodRx and Drugs.com, US list prices (not insurance-adjusted)
- Full SQL queries, source data, and dashboard available in the accompanying GitHub repository and Power BI file